JOINS Fundamentals

1. INNER JOIN:

This join returns only the rows that have matching values in both tables.

```
SELECT a.column1, b.column2
FROM table_a a
INNER JOIN table_b  b
ON a.id = b.id;
```

This query retrieves rows where the `id` in `table_a` matches the `id` in `table_b`.

2. LEFT (OUTER) JOIN:

This returns all rows from the left table and the matched rows from the right table. If there is no match, NULLs are returned for columns from the right table.

```
SELECT a.column1, b.column2
FROM table_a a
LEFT JOIN table_b b
ON a.id = b.id;
```

This returns all records from `table_a` and the matched records from `table_b`. Rows from `table_a` without a match in `table_b` will show `NULL` for columns from `table_b`.

3. FULL JOIN (or FULL OUTER JOIN):

This returns rows when there is a match in one of the tables. It combines the results of both LEFT and RIGHT joins.

```
SELECT a.column1, b.column2
FROM table_a a
FULL OUTER JOIN table_b b
ON a.id = b.id;
```

This will return all records when there is a match in either table. If there is no match, NULL values are returned for non-matching columns from the other table.

How to achieve FULL OUTER JOIN in MySQL?

The common approach is to combine a LEFT JOIN and a RIGHT JOIN using UNION.

```
SELECT e.emp_name, d.dept_name
FROM Employee e
LEFT JOIN Department d
ON e.dept_id = d.dept_id

UNION

SELECT e.emp_name, d.dept_name
FROM Employee e
RIGHT JOIN Department d
ON e.dept_id = d.dept_id;
```

Employee

emp_id	emp_name	dept_id
1	Amit	10
2	Ravi	20
3	Neha	30

Department

dept_id	dept_name
10	HR
20	Sales
40	Finance

Full Outer Join Result

emp_name	dept_name
Amit	HR
Ravi	Sales
Neha	NULL
NULL	Finance

The `LEFT JOIN` gives all employees, including Neha.

The `RIGHT JOIN` gives all departments, including Finance.

`UNION` combines both results and removes duplicate rows.